

Thetanuts MUBA Hackathon

On-chain options from first principles — then the SDK, the tools, and what it takes to ship in either track.

TRACKS

Two · 1000 USDC each

WHERE IT RUNS

Base mainnet · chainId 8453

What is DeFi?

Not just a new currency — an entire alternative financial system.

01

Then people got creative.

Developers realised you can program more than “Alice pays Bob.” Smart contracts kicked off an explosion: DeFi, NFTs, DAOs, entirely new chains.

02

DeFi is the evolution of that idea.

Lending, borrowing, trading, earning yield — all on-chain, all without a bank.

03

Today: think different app stores.

Ethereum, Solana, Base, Arbitrum — like Apple Store vs Google Play. Same concept, different ecosystems, different rules.

Why DeFi Matters.

If I had to break it into one point — it removes the middleman.

HONEST CAVEAT

The veneer of decentralisation is still weak. You're relying on the devs who wrote the code. Better — not perfect.

01

Stablecoins

Ukraine, Venezuela — people in broken economies use USDC to store value and send money across borders without 5–10% remittance fees.

02

Lending platforms

No months of due diligence, credit checks and paperwork. Collateralise assets on-chain, get capital within minutes.

03

Trading platforms (DEXes)

Markets made from collective liquidity — no listing fees, no gatekeepers. Anyone can create and trade a market.

What are options?

01 • THE RIGHT

A right, not an obligation.

A call is the right to buy at a set price by a set date. A put is the right to sell. You choose whether to use it.

02 • THE PRICE

The premium is what you pay.

It's the cost of that right — and as a buyer it's the most you can lose, no matter where the market goes.

03 • THE SETTLEMENT

On-chain it settles in cash.

No delivery, no margin calls. The contract pays the difference at expiry, and every position is 100% collateralised.

STRIKE

The agreed price

EXPIRY

When the right ends

PREMIUM

What you pay for it

NOTIONAL

The size it controls

Why options matter.

\$44T+
TRADFI DAILY NOTIONAL

< 1%
CRYPTO OPTIONS PENETRATION

01

Largest financial market in the world.

Options dwarf spot and futures in TradFi notional volume — and they're barely touched in crypto.

02

Defined risk.

Your maximum loss is known upfront — the premium you paid. Unlike futures or leveraged spot, you can never lose more than you put in.

03

In crypto: infinitely programmable.

Strategies that would be impossible in TradFi. Hedge, speculate, generate yield — on-chain, 24/7, fully automated.

Two tracks.

Both run on the same live protocol on Base mainnet. Both are judged the same way. The difference is what you're pointing the SDK at.

TRACK 01

Best product built on the Thetanuts SDK

Structured products, vaults, consumer trading apps, options-powered lending, analytics — or something we haven't thought of.

1000 USDC

TRACK 02

AI × Options — an agent that trades

Natural-language trading, an AI strategy or risk copilot, an autonomous hedging agent — placing real trades on OptionBook or OptionFactory.

1000 USDC

Best product on the SDK.

Bring us a real, working product that uses on-chain options in a meaningful way. Deliberately open-ended.

THE ONE HARD RULE

If it would work identically with the Thetanuts calls stubbed out, it isn't really using on-chain options.

DIRECTIONS PEOPLE TAKE

Structured products

You bundle options into one product with a promise a normal person understands — “keep your ETH upside, but you can't lose more than 5%.” They buy one thing; the options behind it do the work.

Vaults

A pot anyone can deposit into. Your code runs an options strategy on a schedule — selling covered calls each week, say — and the yield goes back to depositors. They never touch an option.

Consumer trading apps

Make buying an option feel like using a normal app.

Options-powered lending

Use defined-risk payoffs to change what collateral can do.

Analytics

Read the live book and tell people something they can act on.

Something we haven't thought of

Genuinely the most interesting entries every time.

AI x options.

Build an agent that places a real on-chain options trade through OptionBook or OptionFactory, live on Base mainnet.

01

Natural-language trading

“Hedge my ETH into Friday” becomes a signed fill.

02

Strategy or risk copilot

Reads the book, explains the trade-off, then executes.

03

Autonomous hedging

Watches a position and rolls protection on its own.

START HERE

`npx -y @thetanuts-finance/mcp`

Gives your coding agent the whole SDK as context.

`npm i -g @thetanuts-finance/cli`

Quote and fill from the terminal to sanity-check before you wire it up.

WORTH
KNOWING

Nothing stops one entry taking both tracks. If we're happy with it, you can win both.

How we judge.

We don't score complexity, and we don't score your tech stack. We score the idea — and whether it could survive contact with real users.

01

Does it work?

A real, running product — not a mockup, not a README describing what you would have built.

02

Would anyone actually use it?

Who is it for, why would they pick it over what exists, and does it scale? A couple of honest sentences beats a business plan.

AFTER THE
DEMO

If you plan to keep building it, tell us. We support teams after the hackathon — serious builders only.

The toolkit.

Every tool is on one page. Pick the surface that fits how you're building — they all talk to the same live protocol.

SDK

TypeScript client

Orders, RFQs, fills, positions, pricing. Takes any ethers provider.

```
npm i @thetanuts-finance/thetanuts-client
```

CLI

Terminal access

Wallets, quotes and fills without writing code. `--dry-run` on every command.

```
npm i -g @thetanuts-finance/cli
```

MCP

Context for your agent

`get_sdk_context` loads every module, type and gotcha. One question beats the whole GitBook.

```
npx -y @thetanuts-finance/mcp
```

DOCS

Source of truth

If the slides and the repo disagree, the repo wins. Feed agents `llms-full.txt`.

```
docs.thetanuts.finance/for-builders/sdk
```

LIVE REFERENCE

odette.fi

A daily options platform on Thetanuts infra — 1–3 day expiries, up to 2000× leverage, no liquidation risk. Worth a look to see what a production integration can do. Not a template, and not something to copy for either track.

Get an **RPC** first.

The public Base endpoint works — until your agent polls in a loop or your indexer backfills. Then you lose an hour to mystery timeouts that look like bugs in your code.

GRAB A KEY BEFORE YOU WRITE ANYTHING

Free tier is more than enough for a hackathon.

Alchemy

alchemy.com → create app
→ Base Mainnet

Infura

infura.io → create key →
Base

THEN USE IT EVERYWHERE

```
export THETANUTS_RPC_URL="https://base-mainnet.g.alchemy.com/v2/YOUR_KEY"
```

The SDK takes any ethers provider; the CLI reads **THETANUTS_RPC_URL** or **--rpc-url**.

The 30-second check.

If this prints live prices, you're connected to the real protocol. Everything past it — filling, RFQs, positions, pricing — is in the docs and the repo README.

```
const client = new ThetanutsClient({
  chainId: 8453,
  provider: new ethers.JsonRpcProvider(
    process.env.THETANUTS_RPC_URL
  ),
});
console.log(
  (await client.api.fetchOrders()).length, 'live orders'
);
console.log(await client.api.getMarketData());
```

• MAINNET • REAL FUNDS • READ THIS BEFORE YOU TRADE

Trade small. We mean it.

1–3 USDC covers you. A 1 USDC fill scores exactly the same as 100.

Use a fresh wallet.

Never point an agent at a wallet holding anything you'd miss. Keep cents of ETH for gas.

Never commit a key.

Use THETANUTS_PRIVATE_KEY or a .env — and check your git history before you submit.

Dry-run first.

--dry-run is global on every CLI command. Approve the exact amount, never MaxUint256.

OptionBook or RFQ.

Two ways to get an option. Both produce identical positions on the same cash-settled implementation contracts — the difference is only how the quote reaches you.

OPTIONBOOK

Fill what's already on the book

Signed orders sitting live, ready to take. Read the book, pick the order, fill it.

RFQ

Ask for the one you want

Request a quote for a strike and expiry that isn't on the book, and get priced in real time.

RULE OF THUMB

CHAIN	Base mainnet • chainId 8453
RPC	Free Alchemy or Infura key – not the public endpoint
SDK	npm i @thetanuts-finance/thetanuts-client ethers
CLI	npm i -g @thetanuts-finance/cli • binary: thetanuts
MCP	npx -y @thetanuts-finance/mcp
TOOLS	app.thetanuts.finance/tools
DOCS	docs.thetanuts.finance/for-builders/sdk
REPO	github.com/Thetanuts-Finance/thetanuts-sdk

Where to shout.

TELEGRAM

@ShawnSeanC

Anything blocking you.

DISCORD

Thetanuts chatroom

Inside the MUBA Hackathon server.

GITHUB ISSUES

Bugs and missing docs

File it — we read them. Fastest path to a fix this weekend.

T H A T ' S A W R A P

Now go **build.**

BEFORE YOU LEAVE THE ROOM

- 01 Get a free RPC key
- 02 Run the 30-second check
- 03 Install the MCP server
- 04 Fund a burner wallet on Base
- 05 Pick your track